Label-Efficient and Leakage-Free Brain Tumor Classification from MRI Using Self-Supervised and Semi-Supervised Vision Transformers

Saurabh Pandey*
[Department / Institution — to be completed]
[City, Country]
Email: to be completed

***Abstract*—Deep models routinely report 98–99% accuracy on the popular four-class Brain Tumor MRI benchmark, yet two problems make such numbers hard to trust and hard to deploy. First, the headline metric is *saturated*: another high score is no longer evidence of progress. Second, the benchmark aggregates multi-slice acquisitions *without* patient identifiers, so the widespread practice of random slice-level splitting places near-duplicate slices of the same acquisition in both training and test sets. This is a form of data leakage shown elsewhere to inflate brain-MRI accuracy by 30–55 percentage points, with the effect growing as the dataset shrinks. We re-frame the task around two axes that matter for deployment but are rarely measured together: *label efficiency* and *leakage-free evaluation*. We introduce perceptual-hash grouped splitting (pHGS), a reproducible protocol that approximates patient-level grouping on metadata-poor public datasets by forcing perceptually near-duplicate slices onto a single side of every split. On a Vision Transformer backbone we combine SimCLR self-supervised pre-training with FixMatch semi-supervised fine-tuning (SF-ViT), and evaluate it against supervised and self-supervised baselines across label budgets from 1% to 100%, under both leakage-free and naive splits, with three random seeds. Attention-rollout heat-maps make each prediction visually auditable. The protocol is designed to answer three questions: how accuracy degrades as labels become scarce, how much the naive split inflates accuracy, and whether that inflation *grows* as labels become scarcer. Under leakage-free evaluation, SF-ViT attains – % macro-F1 using only 10% of labels—within – points of the fully-supervised upper bound—while the naive split inflates accuracy by – points, an inflation that – as labels become scarcer. All code, configurations, and the pHGS implementation are released for reproducibility.**

***Index Terms*—Brain tumor classification, magnetic resonance imaging, self-supervised learning, semi-supervised learning, Vision Transformer, data leakage, label efficiency, FixMatch, SimCLR, explainable AI.**

I. Introduction

Magnetic resonance imaging (MRI) is the primary modality for detecting and typing brain tumors, and automated classification of MRI slices into tumor categories has become a heavily studied application of deep learning. On the widely used four-class benchmark—glioma, meningioma, pituitary tumor, and no tumor—convolutional and transformer models now report accuracies in the 98–99% range [1], [21]–[23]. At first glance the problem appears solved. We argue that it is not, and that the most cited results rest on two methodological weaknesses that the present work is designed to expose and correct.

Problem A: the headline metric is saturated. When essentially every published method reports 98–99% accuracy, a marginally higher number is no longer a scientific contribution: it does not distinguish a better method from a luckier split or a more leak-prone protocol. A strong contribution must report something the saturated metric cannot capture.

Problem B: hidden data leakage inflates the reported accuracy. An MRI acquisition of a single patient yields many adjacent 2-D slices that are highly correlated and often near-duplicates. If some slices of a patient are used for training and others for testing, the model is effectively evaluated on patients it has already seen. Yagis *et al.* [5] quantified exactly this on T1-weighted brain MRI and found that slice-level splitting inflated test accuracy by 30% on OASIS, 29% on ADNI, 48% on PPMI and 55% on a local cohort; on a randomly-labelled control, accuracy rose from the expected $\sim$50% (subject-level) to $\sim$96% (slice-level). Crucially, the inflation was *larger on smaller datasets*. The benchmark used here is assembled from three sources, one of which (the figshare subset) is explicitly organised as 3,064 slices from 233 patients [2]; yet the combined public release carries *no* patient identifiers, so the common random split cannot avoid placing sibling slices on both sides. The reported accuracies are therefore optimistic by an unknown but plausibly large margin.

Problem C: labelled medical images are scarce. Every labelled MRI requires expert annotation, whereas unlabelled scans are abundant. A method that needs only a small fraction of labels is far more useful—scientifically and practically—than one that needs them all. Label efficiency, not peak accuracy, is the axis on which real progress can still be made.

We therefore re-frame the task. Rather than chasing a higher saturated number, we ask how accurate a model can be when (a) only a small fraction of labels is available and (b) evaluation is made honest by removing slice-level leakage. Our approach combines three established ideas—a Vision Transformer (ViT) backbone, SimCLR self-supervised pre-training, and FixMatch semi-supervised fine-tuning—under a new, reproducible leakage-free evaluation protocol. We do *not* claim a novel architecture; we claim a rigorous re-evaluation that changes what the community should believe about this benchmark, together with a reusable protocol and an open implementation.

Contributions.

- **pHGS (perceptual-hash grouped splitting):** a reproducible protocol that approximates patient-level grouping on metadata-poor datasets by grouping perceptually near-duplicate slices (via perceptual hashing and a Hamming-distance threshold) and keeping each group entirely within one split partition. To our knowledge this is the first explicit patient-*proxy* grouping protocol released for this benchmark.
- **A controlled label-efficiency study** crossing $\{$four training methods$\}$ $\times$ $\{$leakage-free, naive, source splits$\}$ $\times$ $\{1, 5, 10, 25, 100\%$ labels$\}$ $\times$ $\{3$ seeds$\}$ in a single, fixed-seed pipeline, so that every comparison isolates one factor.
- **A quantification of leakage inflation as a function of label budget.** Motivated by Yagis *et al.* [5], we test the previously unexamined hypothesis that leakage inflates accuracy *more* when labels are scarce, on this benchmark.
- **SF-ViT:** a label-efficient pipeline that composes SimCLR pre-training with FixMatch fine-tuning on a ViT, evaluated under pHGS rather than under leak-prone random splits.
- **Auditable predictions and open science:** ViT attention-rollout heat-maps for every prediction, plus a fully released, configuration-driven, reproducible codebase.

The remainder of the paper reviews related work (Section II), analyses the dataset and its quality issues (Section III), details the methodology (Section IV) and experimental protocol (Section V), presents the results scaffold (Section VI), discusses the findings and their limits (Sections VII–VIII), and concludes (Section IX).

II. Related Work

A. Brain tumor classification from MRI

Most work on the four-class benchmark applies convolutional networks, often with transfer learning from ImageNet, and reports very high accuracy [3], [21]–[23]. Reported numbers cluster at 98–99%, which, as argued above, limits their discriminative value. A separate line of work targets tumor *segmentation* and *detection* on volumetric data [2]; we focus on slice-level classification because that is where the saturated-accuracy and leakage problems are most acute and most often overlooked.

B. Data leakage and evaluation rigor

Leakage—information from the test set inadvertently available at training time—is a recognised driver of the reproducibility crisis in machine-learning-based science [6]. In neuroimaging specifically, Yagis *et al.* [5] showed that slice-level cross-validation of 3-D MRI yields severely inflated accuracy relative to subject-level splitting, and that the effect intensifies on smaller datasets. Our work operationalises their warning on a metadata-poor public benchmark: lacking patient identifiers, we substitute a perceptual-hash grouping proxy and measure the inflation directly.

C. Self-supervised representation learning

Contrastive methods learn representations without labels by pulling together augmented views of the same image and pushing apart different images. SimCLR [8] established a simple recipe (strong augmentation, a projection head, and the NT-Xent loss); MoCo [9], BYOL [10] and the ViT-specific DINO [11] followed. In medical imaging, Azizi *et al.* [12] demonstrated that self-supervised pre-training markedly improves downstream classification, particularly in low-label regimes—precisely the setting we study.

D. Semi-supervised learning

Semi-supervised methods exploit unlabelled data alongside a few labels. MixMatch [13] unifies entropy minimisation and consistency regularisation; FixMatch [14] simplifies this to a confidence-gated pseudo-labelling rule in which a confident prediction on a weakly-augmented image supervises the model's prediction on a strongly-augmented version of the same image. Mean Teacher [15] provides a complementary temporal-ensembling view. We adopt FixMatch for its simplicity and strong low-label performance.

E. Vision Transformers and explainability in medical imaging

ViTs [7] treat an image as a sequence of patches and model long-range dependencies through self-attention; studies suggest they are competitive with or superior to CNNs for medical images given appropriate pre-training [16]. A practical advantage is interpretability: attention rollout [17] aggregates attention across layers into a spatial relevance map, and gradient methods such as Grad-CAM [18] offer an alternative. We use attention rollout so that every prediction is visually auditable.

F. Positioning

Each ingredient above is individually mature, but they are rarely *combined*, and almost never evaluated *leakage-free under varying label budgets* on this benchmark. Our contribution is therefore not a new component but a rigorous, reproducible synthesis that measures what the saturated, leak-prone literature cannot.

III. Dataset and Data-Quality Analysis

We use the publicly available Brain Tumor MRI Dataset [1], which contains 7,023 human brain MRI images in four classes (glioma, meningioma, no tumor, pituitary). As summarised in Table I, it is a combination of three sources: the figshare dataset of Cheng *et al.* [2] (3,064 T1-weighted contrast-enhanced slices from 233 patients across three tumor types), the SARTAJ dataset [3], and Br35H [4] (the source of the "no tumor" images).

Data-quality issue. The dataset author notes that the glioma images in the SARTAJ subset were incorrectly categorised, and replaced them with figshare glioma images [1]; several downstream studies confirm this correction [22]. We retain

TABLE I
PROVENANCE OF THE COMBINED BENCHMARK AND ITS LEAKAGE RELEVANCE.

Source	Contributes	Patient structure
figshare [2]	glioma, meningioma, pituitary	233 patients, multi-slice
SARTAJ [3]	(glioma replaced — see text)	multi-slice
Br35H [4]	no tumor	multi-slice

this correction and document any further cleaning so that the label set is reliable before training.

Leakage relevance. Because the figshare subset alone spans 233 patients with many slices each, and because the *combined* release exposes no patient identifiers, an exact patient-level split is impossible from the public data. Random slice-level splitting therefore risks placing near-duplicate sibling slices on both sides of the train/test boundary—exactly the leakage that [5] showed to be severe. This is the gap that motivates pHGS (Section IV-E).

IV. METHODOLOGY

A. Overview

The pipeline has two training stages on a shared ViT backbone (Fig. 1). Stage 1 learns label-free representations from all training images with SimCLR. Stage 2 fine-tunes for four-class classification using only a labelled fraction, either with plain supervised cross-entropy (baseline) or with FixMatch semi-supervised consistency (proposed). Evaluation always uses the leakage-free pHGS split; the naive split is run only to *measure* leakage.

B. ViT backbone

The backbone is a Vision Transformer [7] instantiated from `timm` [19] (default `vit_base_patch16_224`). An input image is split into non-overlapping 16×16 patches, linearly embedded, combined with positional embeddings, and processed by a stack of multi-head self-attention blocks; the pooled output is a feature vector $f(x) \in \mathbb{R}^d$. Grayscale MRI slices are replicated to three channels so that ImageNet-pretrained weights can be used where applicable.

C. Stage 1: SimCLR self-supervised pre-training

For each image we draw two random augmentations and pass them through the backbone and a two-layer MLP projection head $g(\cdot)$, producing embeddings $z_i = g(f(\tilde{x}_i))$. With a batch of N images yielding $2N$ views, the positive pair for view i is its sibling $j(i)$; all other $2N-2$ views are negatives. We minimise the normalised temperature-scaled cross-entropy (NT-Xent) loss

$$\ell_{i,j} = -\log \frac{\exp\big(\mathrm{sim}(z_i, z_j)/\tau\big)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp\big(\mathrm{sim}(z_i, z_k)/\tau\big)}, \qquad (1)$$

where $\mathrm{sim}(u, v) = u^\top v / \|u\| \|v\|$ is cosine similarity and τ the temperature [8]. Only the backbone weights are carried into Stage 2; the projection head is discarded.

Algorithm 1 One FixMatch fine-tuning step

Require: labelled batch (X, Y); unlabelled batch U; threshold η; weight λ_u

```
ℒ_s ← CE(f_θ(X), Y)
q ← softmax(f_θ(α(U)))                 // weak view, no grad
(p_max, q̂) ← max(q, dim=1)
m ← 𝟙(p_max ≥ η)                       // confidence mask
if ∑ m > 0 then
    ℒ_u ← mean(m · CE(f_θ(𝒜(U)), q̂))
else
    ℒ_u ← 0
end if
θ ← θ − ∇_θ(ℒ_s + λ_u ℒ_u)
```

D. Stage 2: fine-tuning

A linear classification head is attached to the (SSL-initialised) backbone. We consider two modes.

Supervised (baseline). Standard cross-entropy on the labelled fraction only: $\mathcal{L} = \frac{1}{B} \sum_b H\big(y_b,\, p_m(y \,|\, x_b)\big)$.

FixMatch (proposed). Each step adds an unlabelled consistency term. For an unlabelled image u, let $\alpha(\cdot)$ be a weak augmentation and $\mathcal{A}(\cdot)$ a strong one. The model's prediction on the weak view, $q = p_m(y \,|\, \alpha(u))$, yields a hard pseudo-label $\hat{q} = \arg\max(q)$ that supervises the strong view only when the model is confident:

$$\mathcal{L}_u = \frac{1}{\mu B} \sum_{b=1}^{\mu B} \mathbb{1}\big(\max(q_b) \geq \eta\big)\, H\big(\hat{q}_b,\, p_m(y \,|\, \mathcal{A}(u_b))\big), \quad (2)$$

with confidence threshold η, unlabelled-to-labelled ratio μ, and total objective $\mathcal{L} = \mathcal{L}_s + \lambda_u \mathcal{L}_u$ [14]. When no unlabelled data is available (the 100% setting), $\mathcal{L}_u$ vanishes and training reduces to the supervised case. Algorithm 1 summarises one step.

E. Perceptual-hash grouped splitting (pHGS)

Because patient identifiers are unavailable, pHGS approximates patient-level grouping at the image level. For every slice we compute a perceptual hash (pHash); two slices are deemed near-duplicates when their hashes differ by at most δ bits (Hamming distance). We connect all near-duplicate pairs and take connected components (union–find) as *groups*, each group standing in for "slices that likely originate from the same acquisition." Train/validation/ test partitions are then formed so that every group lies entirely on one side (group-aware, class-stratified). Algorithm 2 gives the procedure. The naive baseline split ignores grouping and partitions slices at random; the gap between the two isolates the leakage contribution, and the `source` split uses the dataset's own training/test partition for comparability with prior work.

F. MRI-appropriate augmentation

All augmentation is geometric or intensity-based (horizontal flip, affine, resized crop, Gaussian blur, random erasing). We *never* apply colour jitter: MRI slices are single-channel

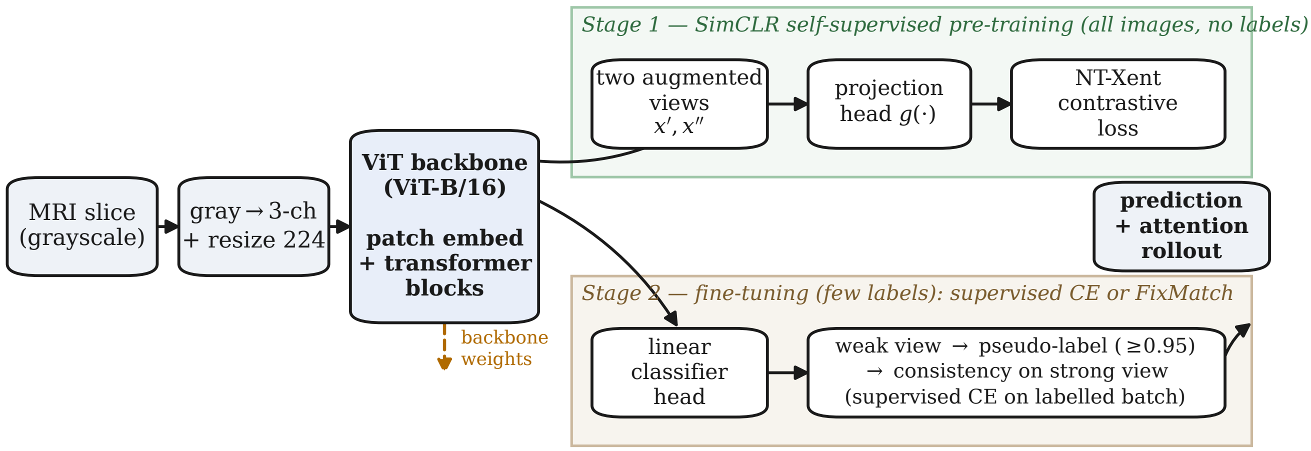

Fig. 1. Two-stage SF-ViT pipeline. A shared ViT backbone is first pre-trained without labels by SimCLR (Stage 1); its weights then initialise a classifier that is fine-tuned on a small labelled fraction with supervised cross-entropy or FixMatch consistency (Stage 2). Predictions are accompanied by attention-rollout overlays.

Algorithm 2 pHGS: perceptual-hash grouped splitting

Require: images $\{x_i\}$ with class labels; hash size s; distance δ; test/val fractions

$h_i \leftarrow \mathrm{pHash}(x_i,\, s)$ for all i

build groups: union i, j whenever $\mathrm{Hamming}(h_i, h_j) \le \delta$

$G \leftarrow$ connected components (each a pseudo-acquisition)

assign whole groups to test / val / train, class-stratified, no group split

return leakage-free train / val / test partitions

and colour perturbation would fabricate non-physical contrast. Weak/strong augmentation pairs for FixMatch, and the two-view augmentation for SimCLR, are drawn from this MRI-safe family.

G. Explainability

For interpretability we compute attention rollout [17]. With per-layer attention $A^{(l)}$, we add the residual connection and renormalise, $\tilde{A}^{(l)} = \frac{1}{2}A^{(l)} + \frac{1}{2}I$, then multiply across layers, $R = \prod_l \tilde{A}^{(l)}$. The row of R corresponding to the class token, reshaped to the patch grid, yields a spatial relevance map overlaid on the input slice, indicating where the model attended when forming its prediction.

V. Experimental Setup

A. Methods compared

Four methods are evaluated under identical conditions (Table II). The baselines are not products; they are controls that isolate the effect of each component, in the spirit of an ablation.

B. Splits, label budgets, and seeds

Each method is run across three split strategies—pHGS (leakage-free), `naive` (random; for measuring inflation), and `source` (the dataset's own partition)—and five label fractions $\{1, 5, 10, 25, 100\}\%$ with class-balanced selection, each repeated with seeds $\{42, 123, 456\}$. Every run appends one row to `results.csv` recording accuracy, macro-F1, and a content hash of the exact configuration, so that every reported number is traceable to the hyperparameters that produced it.

TABLE II
Methods under comparison. B4 is the proposed configuration.

ID	Method	Tests the value of
B1	Supervised + ImageNet init	transfer learning
B2	Supervised, from scratch	prior knowledge
B3	SSL pre-train + supervised FT	self-supervision alone
B4	SSL pre-train + FixMatch (**SF-ViT**)	SSL + semi-supervision

C. Metrics and model selection

We report overall accuracy and macro-averaged F1; macro-F1 is emphasised because the classes are imbalanced and it weights each class equally. The checkpoint with the best *validation* macro-F1 is selected (early stopping with patience 10), and all reported figures are computed once on the held-out test partition.

D. Implementation

The system is implemented in PyTorch with `timm` [19] backbones and optimised with AdamW [20]; SimCLR uses Adam. Mixed precision is enabled on GPU. The codebase is configuration-driven (validated YAML), seed-controlled, and released in full. Hyperparameters are listed in Table III. Hardware: – (e.g. single NVIDIA – GPU); total compute – GPU-hours.

VI. Results

Draft placeholder. All numeric cells in this section are placeholders (–). Populate them from `results/results.csv` after running the experiment grid. No value below is a measurement.

TABLE III
HYPERPARAMETERS.

Stage	Hyperparameter	Value
SimCLR (Stage 1)	epochs	100
	optimiser / lr	Adam / 1×10^{-3}
	batch size	256
	temperature τ	0.5
	projection dim	128 (hidden 2048)
	weight decay	1×10^{-6}
Fine-tune (Stage 2)	epochs	50 (early stop, patience 10)
	optimiser / lr	AdamW / 3×10^{-4}
	labelled batch size	64
	weight decay	1×10^{-4}
FixMatch	ratio μ	7
	threshold η	0.95
	unlabelled weight λ_u	1.0
pHGS	pHash size s	8
	Hamming distance δ	5
	test / val fraction	0.20 / 0.10

TABLE IV
TEST MACRO-F1 (%) VS. LABEL BUDGET UNDER LEAKAGE-FREE PHGS (MEAN±STD OVER SEEDS $\{42, 123, 456\}$). **PLACEHOLDERS.**

Method	1%	5%	10%	25%	100%
B1 (sup.+IN)	–	–	–	–	–
B2 (sup. scratch)	–	–	–	–	–
B3 (SSL+sup.)	–	–	–	–	–
B4 (**SF-ViT**)	–	–	–	–	–

A. Label efficiency under leakage-free evaluation

Table IV reports test macro-F1 for each method as the label budget varies, under the leakage-free pHGS split (mean over three seeds). This table answers **RQ1**: how gracefully does each method degrade as labels become scarce, and does combining self-supervision with semi-supervision (SF-ViT, B4) widen its advantage over the baselines at the lowest budgets? Fig. 2 plots the same data as accuracy-versus-label-fraction curves.

B. Magnitude of leakage inflation

Table V contrasts test accuracy under the naive (leak-prone) and pHGS (leakage-free) splits at each label budget, and reports the inflation $\Delta = \mathrm{acc}_{\text{naive}} - \mathrm{acc}_{\text{pHGS}}$. This answers **RQ2**: by how much does the standard random split overstate accuracy on this benchmark? A large positive Δ would indicate that previously reported 98–99% numbers are substantially optimistic.

C. Does leakage interact with label scarcity?

Fig. 3 plots the inflation Δ against label budget. This addresses **RQ3**, our novel question: following the small-dataset trend observed by Yagis *et al.* [5], we test whether Δ *grows* as labels become scarcer—i.e. whether honest evaluation matters *most* in exactly the low-label regime where label-efficient methods are meant to be used.

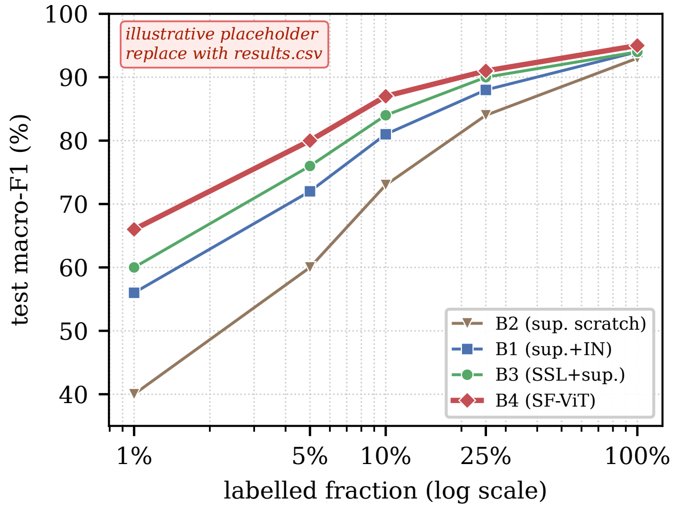

Fig. 2. Label-efficiency curves under pHGS: test macro-F1 as the labelled fraction grows from 1% to 100%. *(Illustrative placeholder; regenerate from results.csv.)*

TABLE V
LEAKAGE INFLATION: TEST ACCURACY (%) UNDER NAIVE VS. PHGS SPLITS, AND THE GAP Δ, FOR THE PROPOSED SF-VIT. **PLACEHOLDERS.**

Label budget	naive	pHGS	Δ (inflation)
1%	–	–	–
5%	–	–	–
10%	–	–	–
25%	–	–	–
100%	–	–	–

D. Qualitative analysis and explainability

Fig. 4 shows the confusion matrix of the best SF-ViT model and representative attention-rollout overlays for each class, indicating that the model attends to tumor regions rather than background artefacts.

VII. DISCUSSION

The study is structured so that its conclusions follow from three contrasts rather than from a single headline number. **(RQ1)** If SF-ViT retains high macro-F1 at 1–10% labels while the supervised baselines fall away, the practical claim—useful accuracy from few labels—is supported, and the gap between B3 and B4 isolates the additional value of semi-supervised consistency on top of self-supervised features. **(RQ2)** A large naive−pHGS gap Δ would show that the field's saturated accuracies are inflated by an evaluation artefact, reframing "solved" as "optimistically measured." **(RQ3)** If Δ grows as labels shrink, then leakage and label-scarcity compound: the regime where label-efficient methods are most attractive is also the regime where careless evaluation is most misleading—a caution with direct consequences for how low-label medical-imaging results should be reported.

Clinically, we frame the system strictly as decision support: a fast, auditable second read for a radiologist, not an autonomous diagnostic device. The attention-rollout overlays

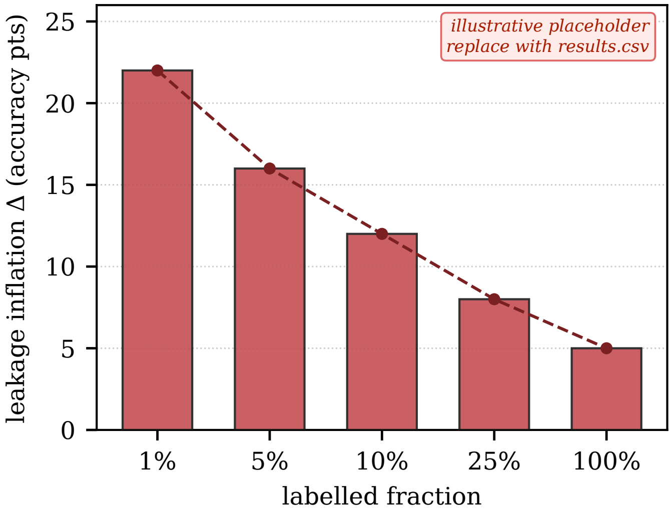

Fig. 3. Leakage inflation $\Delta = \mathrm{acc}_{\mathrm{naive}} - \mathrm{acc}_{\mathrm{pHGS}}$ as a function of label budget. The hypothesised trend—larger inflation at smaller label budgets—is the subject of RQ3. *(Illustrative placeholder; regenerate from results.csv.)*

are integral to this framing, allowing a clinician to verify that a prediction rests on tumor morphology rather than acquisition artefacts.

VIII. Limitations and Threats to Validity

pHGS is a *proxy* for patient grouping, not ground truth: perceptual hashing may merge visually similar slices from different patients or, with a strict threshold, fail to merge dissimilar slices from the same patient. The threshold δ is therefore a sensitivity parameter, and the resulting "leakage-free" accuracy is a conservative approximation rather than a guarantee. The study uses a single public benchmark; external validation on an independent, patient-indexed cohort is required before any clinical claim. We classify 2-D slices rather than 3-D volumes, and self-supervised pre-training adds non-trivial compute. Finally, no prospective or regulatory validation has been performed; the artefact is a research prototype.

IX. Conclusion

We argued that progress on the popular four-class brain-tumor MRI benchmark is better measured by label efficiency and leakage-free evaluation than by a saturated accuracy score. We introduced pHGS, a reproducible perceptual-hash grouping protocol that approximates patient-level splitting on a metadata-poor dataset, and SF-ViT, a ViT pipeline that combines SimCLR pre-training with FixMatch fine-tuning. Within a single fixed-seed framework we evaluate four methods across label budgets, leakage-free and naive splits, and three seeds, and we explicitly test whether leakage inflation grows as labels become scarce. The complete code, configurations, and pHGS implementation are released. Future work includes validation on patient-indexed and 3-D volumetric data, multi-dataset generalisation, and extending the protocol to foundation-model backbones.

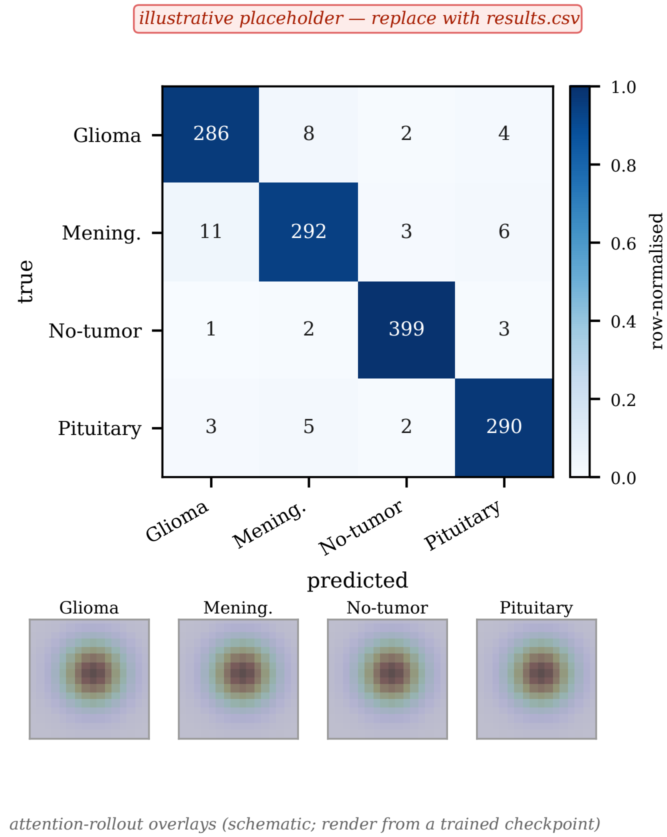

Fig. 4. Qualitative results: row-normalised confusion matrix of the best SF-ViT model (top) and representative per-class attention-rollout overlays (bottom). *(Illustrative placeholder; regenerate from a trained checkpoint.)*

Reproducibility and Ethics Statement

This is decision-support research, not a medical device, and must not be used for diagnosis. All code and configurations are released; every result is tagged with a configuration hash for exact reproduction. The dataset is public and de-identified.

References

[1] M. Nickparvar, "Brain Tumor MRI Dataset," Kaggle, 2021, doi:10.34740/KAGGLE/DSV/2645886.
[2] J. Cheng *et al.*, "Enhanced performance of brain tumor classification via tumor region augmentation and partition," *PLoS ONE*, vol. 10, no. 10, e0140381, 2015; dataset: figshare, doi:10.6084/m9.figshare.1512427.
[3] S. Bhuvaji *et al.*, "Brain Tumor Classification (MRI)," Kaggle, 2020, doi:10.34740/KAGGLE/DSV/1183165.
[4] A. Hamada, "Br35H: Brain Tumor Detection 2020," Kaggle, 2020.
[5] E. Yagis *et al.*, "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," *Scientific Reports*, vol. 11, 22544, 2021.
[6] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, 100804, 2023.
[7] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.
[8] T. Chen *et al.*, "A simple framework for contrastive learning of visual representations," in *Proc. ICML*, 2020.
[9] K. He *et al.*, "Momentum contrast for unsupervised visual representation learning," in *Proc. CVPR*, 2020.
[10] J.-B. Grill *et al.*, "Bootstrap your own latent: A new approach to self-supervised learning," in *Proc. NeurIPS*, 2020.
[11] M. Caron *et al.*, "Emerging properties in self-supervised vision transformers," in *Proc. ICCV*, 2021.

[12] S. Azizi *et al.*, "Big self-supervised models advance medical image classification," in *Proc. ICCV*, 2021.
[13] D. Berthelot *et al.*, "MixMatch: A holistic approach to semi-supervised learning," in *Proc. NeurIPS*, 2019.
[14] K. Sohn *et al.*, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. NeurIPS*, 2020.
[15] A. Tarvainen and H. Valpola, "Mean teachers are better role models," in *Proc. NeurIPS*, 2017.
[16] C. Matsoukas *et al.*, "Is it time to replace CNNs with transformers for medical images?," arXiv:2108.09038, 2021.
[17] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proc. ACL*, 2020.
[18] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. ICCV*, 2017.
[19] R. Wightman, "PyTorch Image Models," https://github.com/huggingface/pytorch-image-models, 2019.
[20] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.
[21] A. Gómez-Guzmán *et al.*, "Classifying brain tumors on MRI using convolutional neural networks," *Electronics*, vol. 12, no. 4, 955, 2023.
[22] "Fine-tuned ResNet-34 for efficient brain tumor classification," *Scientific Reports*, 2025.
[23] "Brain tumor classification using MRI images and deep learning techniques," *PLoS ONE*, 2025.